

Final Project Report:

*Inflation, Energy Prices, and the Strait of Hormuz:
Correlations Under Geopolitical Stress (2019-2025)*

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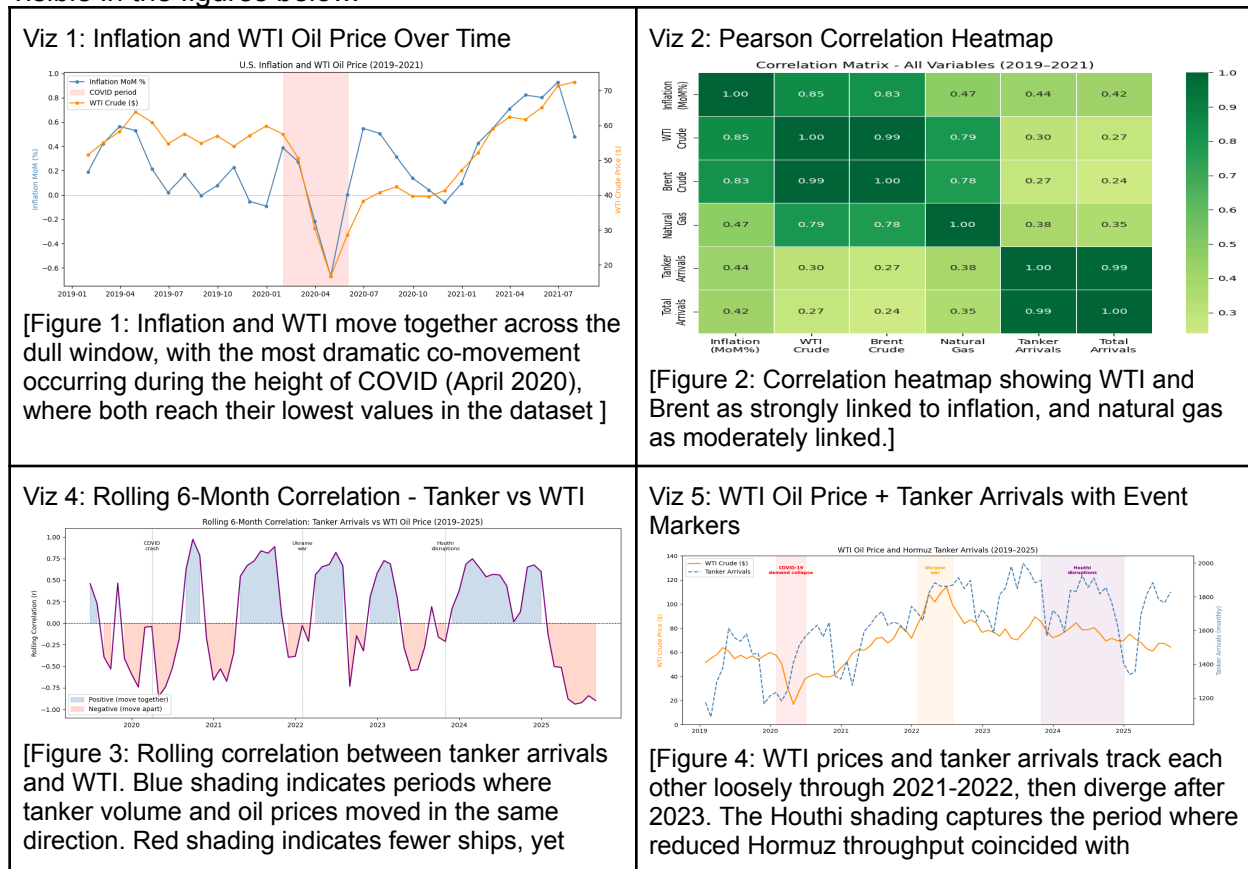
Problem Statement

This project investigates the relationship between U.S. inflation and oil/gas prices, with additional focus on tanker traffic through the Strait of Hormuz as a proxy for global energy supply disruptions. The central research question asks how inflation and energy prices have moved together over time, and whether that relationship shifts during periods of geopolitical stress. The analysis treats the Strait of Hormuz, a critical checkpoint for global oil trade, as a third variable that may help explain price volatility beyond domestic factors alone.

Two scope changes were made relative to the original proposal. The ten-year window was shortened to five years (2020-2025) after reviewing the datasets' actual coverage. More significantly, the analysis uses month-over-month (MoM) inflation rather than year-over-year (YoY) because the YoY measure requires twelve months of prior data, reducing the usable merged window to only seven rows, too few for any meaningful analysis. Switching to MoM preserved all 31 months of the full three-dataset overlap. The project doesn't claim causation; our goal is to identify and describe observable correlations.

Outcomes & Results

The analysis operated across two windows: a full three-dataset merge (CPI, commodity prices, and Hormuz ship arrivals) spanning January 2019 to July 2021, and a longer window between the Resources and ship arrivals datasets extending to August 2025. Three major geopolitical events shaped the data across this period: the COVID-19 pandemic triggered a historic collapse in global oil demand (U.S. Energy Information Administration, 2020); the Russia-Ukraine war (2022) drove WTI to a record high of \$114 (International Energy Agency, 2022); and Houthi attacks on Red Sea and Gulf of Aden shipping (late 2023–present) reduced Hormuz tanker traffic while oil prices remained elevated (IMF PortWatch, 2024). Each event is visible in the figures below.



prices remain elevated. A sustained red region emerges after the Houthi disruptions began.]	sustained high oil prices rather than a corresponding price decline.]
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Oil prices and inflation moved together strongly across the 2019–2021 window, with WTI showing the strongest relationship ($r = 0.85$) and total arrivals the weakest ($r = 0.42$). Brent tracked closely behind WTI ($r = 0.83$), while natural gas and tanker arrivals showed moderate positive relationships ($r = 0.47$ and $r = 0.44$, respectively). The COVID period produced the most extreme values. Most significantly, the Houthi disruptions after 2023 broke the historically positive tanker-oil relationship, revealing that Hormuz throughput matters most as a signal during geopolitical stress. The correlation between energy prices and inflation is real, but not stable across all periods.

Final Deliverable

The final deliverables are a collection of materials, which include original datasets, a separate notebook for data cleaning, an initial analysis, and a final analysis. We also created a notebook focusing on machine learning for predictive analysis as an extra step. These can be used to show how inflation, energy costs, and oil tanker traffic are related in the US. The target audience is those who are concerned about the gas prices and who want to understand how the change happened. We chose to use a notebook format to show the analysis and findings in a reproducible manner. These are available on our shared [Google Drive](#).

Track-specific Reflection

In our project proposal, we claimed that “the problem does not lie in collecting data, but in how one can organize, compare, and interpret the variables in a clear and repeatable process.” To a large extent, this was true. Notebooks 1, 2, and 3 each required substantial cleaning: converting daily commodity and Hormuz data to monthly frequency, standardizing date formats across three differently structured CSVs, and aligning all datasets to a shared month-end join key before any analysis could begin. We didn’t anticipate how little all the datasets would overlap (the CPI data ends in July 2021, limiting the full three-dataset window to 31 months). We also didn’t anticipate the Houthis’ disruptions as an analytical factor; their effect only became visible once the rolling correlation chart revealed a structural break after 2023, forcing us to contextualize a geopolitical event that was not a part of our original research question.

Final Contribution

Joan Ojukwu: Data curation, formal analysis, investigation, methodology, project administration, resources, software (lead), validation, writing (review and editing). Led all data cleaning across Notebooks 1-3, including date standardization and dataset alignment. Drove key methodology decisions (the MoM inflation switch), which shaped comparative analysis.

Duckwoo Kwon: Formal analysis, methodology, project administration, resources, software (assistance), writing(draft). In the proposal, planned to be on data curation and analysis. Initially created the first draft of the analysis notebook. The final version was later led and refined by Joan. Created additional visualizations to support the analysis and wrote a visualization document on our Google Drive.

Michael Brown: Investigation, find resources. In the proposal, planned to focus on data curation and analysis. Primary researcher, notably, the Strait of Hormuz and its impact on US oil prices. Added historical context to explain why we targeted a specific chokepoint and why it is so consequential to the global oil trade.

References

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